Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

ADBMS Coursework Submission

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-linu compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

to see what all colums needed form 'pg_stats'

```
\d pg_stats;
```

to see histogram boundary values and other relevent stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

PSQL uses an Equi Depth Histogram combined with a Most Common Values list. In an equidepth histogram, all buckets contain approximately the same number of rows ($approx N/B$), while bucket boundaries vary in width according to data distribution

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)
- extra - to see the defualt number of buckets

```
SHOW default_statistics_target;
```

3.1.4 4. How Textual Attribute `title` is Put into Histogram

textual attribute "title" is put into histogram as:

1. sorting the text strings in lexicographical order like bannana and apple will be apple, bannana after sorting

2. Sampling string values from the table like table title has around 2.5M rows , it will only sample around 30K

3. Encoding initial string prefix bytes into floating-point numbers in $[0.0, 1.0]$ range for quantitative interpolation like laptop will be broken to lap with l = 108, a=97, p=112 then by 256 as between 0 and 1 , lap will be around 0.42 , we can then see where a key belongs between two keeys

4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equi-depth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From `pg_stats`)

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`id`)	Bucket Interval
B000 (Start)	85	Start of table data
B001	23414	Bucket 1: [85 - 23414]
B002	50842	Bucket 2: [23414 - 50842]
B003	72092	Bucket 3: [50842 - 72092]
B004	95754	Bucket 4: [72092 - 95754]
B005	120981	Bucket 5: [95754 - 120981]
B006	146382	Bucket 6: [120981 - 146382]

B007	169338	Bucket 7: [146382 - 169338]
B008	193625	Bucket 8: [169338 - 193625]
B009	222737	Bucket 9: [193625 - 222737]
B010	248023	Bucket 10: [222737 - 248023]
B011	273293	Bucket 11: [248023 - 273293]
B012	297400	Bucket 12: [273293 - 297400]
B013	321995	Bucket 13: [297400 - 321995]
B014	348628	Bucket 14: [321995 - 348628]
B015	372223	Bucket 15: [348628 - 372223]
B016	399997	Bucket 16: [372223 - 399997]
B017	425563	Bucket 17: [399997 - 425563]
B018	452049	Bucket 18: [425563 - 452049]
B019	477594	Bucket 19: [452049 - 477594]
B020	503120	Bucket 20: [477594 - 503120]
B021	527541	Bucket 21: [503120 - 527541]
B022	550901	Bucket 22: [527541 - 550901]
B023	578985	Bucket 23: [550901 - 578985]
B024	602917	Bucket 24: [578985 - 602917]
B025	629391	Bucket 25: [602917 - 629391]
B026	655601	Bucket 26: [629391 - 655601]
B027	681541	Bucket 27: [655601 - 681541]
B028	705430	Bucket 28: [681541 - 705430]
B029	729765	Bucket 29: [705430 - 729765]
B030	756585	Bucket 30: [729765 - 756585]
B031	784558	Bucket 31: [756585 - 784558]
B032	810790	Bucket 32: [784558 - 810790]
B033	836757	Bucket 33: [810790 - 836757]
B034	861299	Bucket 34: [836757 - 861299]
B035	887193	Bucket 35: [861299 - 887193]
B036	911713	Bucket 36: [887193 - 911713]
B037	936140	Bucket 37: [911713 - 936140]
B038	964425	Bucket 38: [936140 - 964425]
B039	990604	Bucket 39: [964425 - 990604]
B040	1017549	Bucket 40: [990604 - 1017549]
B041	1042518	Bucket 41: [1017549 - 1042518]
B042	1068162	Bucket 42: [1042518 - 1068162]
B043	1092644	Bucket 43: [1068162 - 1092644]
B044	1119041	Bucket 44: [1092644 - 1119041]
B045	1143249	Bucket 45: [1119041 - 1143249]
B046	1167293	Bucket 46: [1143249 - 1167293]
B047	1190899	Bucket 47: [1167293 - 1190899]
B048	1214192	Bucket 48: [1190899 - 1214192]
B049	1237126	Bucket 49: [1214192 - 1237126]
B050	1259630	Bucket 50: [1237126 - 1259630]
B051	1282683	Bucket 51: [1259630 - 1282683]
B052	1307704	Bucket 52: [1282683 - 1307704]
B053	1331946	Bucket 53: [1307704 - 1331946]
B054	1354962	Bucket 54: [1331946 - 1354962]
B055	1381303	Bucket 55: [1354962 - 1381303]
B056	1406118	Bucket 56: [1381303 - 1406118]
B057	1432716	Bucket 57: [1406118 - 1432716]
B058	1460142	Bucket 58: [1432716 - 1460142]
B059	1484816	Bucket 59: [1460142 - 1484816]
B060	1512661	Bucket 60: [1484816 - 1512661]
B061	1538548	Bucket 61: [1512661 - 1538548]
B062	1562535	Bucket 62: [1538548 - 1562535]
B063	1585840	Bucket 63: [1562535 - 1585840]
B064	1611166	Bucket 64: [1585840 - 1611166]
B065	1635920	Bucket 65: [1611166 - 1635920]
B066	1662169	Bucket 66: [1635920 - 1662169]

B067	1688003	Bucket 67: [1662169 - 1688003]
B068	1714390	Bucket 68: [1688003 - 1714390]
B069	1737794	Bucket 69: [1714390 - 1737794]
B070	1762970	Bucket 70: [1737794 - 1762970]
B071	1786873	Bucket 71: [1762970 - 1786873]
B072	1813638	Bucket 72: [1786873 - 1813638]
B073	1840890	Bucket 73: [1813638 - 1840890]
B074	1867370	Bucket 74: [1840890 - 1867370]
B075	1895175	Bucket 75: [1867370 - 1895175]
B076	1919111	Bucket 76: [1895175 - 1919111]
B077	1941477	Bucket 77: [1919111 - 1941477]
B078	1966111	Bucket 78: [1941477 - 1966111]
B079	1992282	Bucket 79: [1966111 - 1992282]
B080	2016500	Bucket 80: [1992282 - 2016500]
B081	2040965	Bucket 81: [2016500 - 2040965]
B082	2067381	Bucket 82: [2040965 - 2067381]
B083	2090687	Bucket 83: [2067381 - 2090687]
B084	2116111	Bucket 84: [2090687 - 2116111]
B085	2142228	Bucket 85: [2116111 - 2142228]
B086	2167352	Bucket 86: [2142228 - 2167352]
B087	2192781	Bucket 87: [2167352 - 2192781]
B088	2217642	Bucket 88: [2192781 - 2217642]
B089	2245252	Bucket 89: [2217642 - 2245252]
B090	2268088	Bucket 90: [2245252 - 2268088]
B091	2292543	Bucket 91: [2268088 - 2292543]
B092	2318646	Bucket 92: [2292543 - 2318646]
B093	2343571	Bucket 93: [2318646 - 2343571]
B094	2369432	Bucket 94: [2343571 - 2369432]
B095	2395899	Bucket 95: [2369432 - 2395899]
B096	2422979	Bucket 96: [2395899 - 2422979]
B097	2448577	Bucket 97: [2422979 - 2448577]
B098	2475828	Bucket 98: [2448577 - 2475828]
B099	2502132	Bucket 99: [2475828 - 2502132]
B100	2528188	Bucket 100: [2502132 - 2528188]

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	`\$100,000 Name That Tune`	Start of text data
B001	`(#1.135)`	Bucket 1: [`\$100,000 Name That Tune` - `(#1.135)`]
B002	`(#1.220)`	Bucket 2: [`(#1.135)` - `(#1.220)`]
B003	`(#1.3630)`	Bucket 3: [`(#1.220)` - `(#1.3630)`]
B004	`(#1.5551)`	Bucket 4: [`(#1.3630)` - `(#1.5551)`]
B005	`(#1.7648)`	Bucket 5: [`(#1.5551)` - `(#1.7648)`]
B006	`(1964-03-09)`	Bucket 6: [`(#1.7648)` - `(1964-03-09)`]
B007	`(#1.981)`	Bucket 7: [`(1964-03-09)` - `(#1.981)`]
B008	`(1993-01-27)`	Bucket 8: [`(#1.981)` - `(1993-01-27)`]
B009	`(1997-10-20)`	Bucket 9: [`(1993-01-27)` - `(1997-10-20)`]
B010	`(2000-01-31)`	Bucket 10: [`(1997-10-20)` - `(2000-01-31)`]
B011	`(2001-10-04)`	Bucket 11: [`(2000-01-31)` - `(2001-10-04)`]
B012	`(2003-08-05)`	Bucket 12: [`(2001-10-04)` - `(2003-08-05)`]
B013	`(2004-11-24)`	Bucket 13: [`(2003-08-05)` - `(2004-11-24)`]
B014	`(2006-03-14)`	Bucket 14: [`(2004-11-24)` - `(2006-03-14)`]
B015	`(2007-06-22)`	Bucket 15: [`(2006-03-14)` - `(2007-06-22)`]
B016	`(2009-01-06)`	Bucket 16: [`(2007-06-22)` - `(2009-01-06)`]
B017	`(2010-10-09)`	Bucket 17: [`(2009-01-06)` - `(2010-10-09)`]
B018	`(2011-10-31)`	Bucket 18: [`(2010-10-09)` - `(2011-10-31)`]
B019	`(2012-11-08)`	Bucket 19: [`(2011-10-31)` - `(2012-11-08)`]

B020	(#2.24)	Bucket 20: [(2012-11-08) - (#2.24)]
B021	(#3.252)	Bucket 21: [(#2.24) - (#3.252)]
B022	(#5.14)	Bucket 22: [(#3.252) - (#5.14)]
B023	(#7.48)	Bucket 23: [(#5.14) - (#7.48)]
B024	According to Jim	Bucket 24: [(#7.48) - According to Jim]
B025	A Girl's Affair 8	Bucket 25: [According to Jim - A Girl's Affair 8]
B026	Älska mej	Bucket 26: [A Girl's Affair 8 - Älska mej]
B027	Animated Stories from the Bible: Music Video - Volume 1	Bucket 27: [Älska mej - Animated Stories from the Bible: Music Video - Volume 1]
B028	As Férias	Bucket 28: [Animated Stories from the Bible: Music Video - Volume 1 - As Férias]
B029	A World Away: Stories from the Regina Five	Bucket 29: [As Férias - A World Away: Stories from the Regina Five]
B030	Beautiful Girls 2	Bucket 30: [A World Away: Stories from the Regina Five - Beautiful Girls 2]
B031	Bill of Mortality	Bucket 31: [Beautiful Girls 2 - Bill of Mortality]
B032	Born for Sex	Bucket 32: [Bill of Mortality - Born for Sex]
B033	Busy Bodies	Bucket 33: [Born for Sex - Busy Bodies]
B034	Cattle Call	Bucket 34: [Busy Bodies - Cattle Call]
B035	Chô-jûryoku no wana	Bucket 35: [Cattle Call - Chô-jûryoku no wana]
B036	Commando Lovers	Bucket 36: [Chô-jûryoku no wana - Commando Lovers]
B037	Cultivating the Desert	Bucket 37: [Commando Lovers - Cultivating the Desert]
B038	Davy Crockett: Rainbow in the Thunder	Bucket 38: [Cultivating the Desert - Davy Crockett: Rainbow in the Thunder]
B039	Der Konzern	Bucket 39: [Davy Crockett: Rainbow in the Thunder - Der Konzern]
B040	Die Reise ins Glück	Bucket 40: [Der Konzern - Die Reise ins Glück]
B041	Dos amigos de cuidado	Bucket 41: [Die Reise ins Glück - Dos amigos de cuidado]
B042	EF	Bucket 42: [Dos amigos de cuidado - EF]
B043	Empire	Bucket 43: [EF - Empire]
B044	Expulsion: Pilot	Bucket 44: [Empire - Expulsion: Pilot]
B045	Finding Balance	Bucket 45: [Expulsion: Pilot - Finding Balance]
B046	Frances Ha	Bucket 46: [Finding Balance - Frances Ha]
B047	Gatten vs. Gatten	Bucket 47: [Frances Ha - Gatten vs. Gatten]
B048	Gone Fishin'	Bucket 48: [Gatten vs. Gatten - Gone Fishin']
B049	Haha no mune ni yomigaeru shonen	Bucket 49: [Gone Fishin' - Haha no mune ni yomigaeru shonen]
B050	Hello Mary Lou: Prom Night II	Bucket 50: [Haha no mune ni yomigaeru shonen - Hello Mary Lou: Prom Night II]
B051	Hollywood on Parade No. B-13	Bucket 51: [Hello Mary Lou: Prom Night II - Hollywood on Parade No. B-13]
B052	Hyakka Ryoran: Samurai Bride	Bucket 52: [Hollywood on Parade No. B-13 - Hyakka Ryoran: Samurai Bride]

B053	Incorrigible	Bucket 53: [Hyakka Ryoran: Samurai Bride - Incorrigible]
B054	Itinéraire de Jean Bricard	Bucket 54: [Incorrigible - Itinéraire de Jean Bricard]
B055	Jimmy Fallon/Kudo Tsunoda/Microsoft Kinect/iPhone 4/Samsung Captivate/Droid X	Bucket 55: [Itinéraire de Jean Bricard - Jimmy Fallon/Kudo Tsunoda/Microsoft Kinect/iPhone 4/Samsung Captivate/Droid X]
B056	Kamoning	Bucket 56: [Jimmy Fallon/Kudo Tsunoda/Microsoft Kinect/iPhone 4/Samsung Captivate/Droid X - Kamoning]
B057	Kishi tachi no hatsujin	Bucket 57: [Kamoning - Kishi tachi no hatsujin]
B058	La duquesa del Tepetate	Bucket 58: [Kishi tachi no hatsujin - La duquesa del Tepetate]
B059	Las sensuales fotos de Sole	Bucket 59: [La duquesa del Tepetate - Las sensuales fotos de Sole]
B060	Le retour des Goraks	Bucket 60: [Las sensuales fotos de Sole - Le retour des Goraks]
B061	Little Buddies	Bucket 61: [Le retour des Goraks - Little Buddies]
B062	Love Hangover	Bucket 62: [Little Buddies - Love Hangover]
B063	Makyo no himitsu	Bucket 63: [Love Hangover - Makyo no himitsu]
B064	Meditations: Supper	Bucket 64: [Makyo no himitsu - Meditations: Supper]
B065	Miss America: Behind the Curtain	Bucket 65: [Meditations: Supper - Miss America: Behind the Curtain]
B066	Mrs. Brickles	Bucket 66: [Miss America: Behind the Curtain - Mrs. Brickles]
B067	Na odmoru	Bucket 67: [Mrs. Brickles - Na odmoru]
B068	Njihova poslednja smjena	Bucket 68: [Na odmoru - Njihova poslednja smjena]
B069	O Dilema	Bucket 69: [Njihova poslednja smjena - O Dilema]
B070	Oppa soll ins Heim	Bucket 70: [O Dilema - Oppa soll ins Heim]
B071	pareidolia	Bucket 71: [Oppa soll ins Heim - pareidolia]
B072	Pick-up	Bucket 72: [pareidolia - Pick-up]
B073	Pozo ciego	Bucket 73: [Pick-up - Pozo ciego]
B074	Querelle enfantine	Bucket 74: [Pozo ciego - Querelle enfantine]
B075	Remember When?	Bucket 75: [Querelle enfantine - Remember When?]
B076	Roger Varley the Sweep	Bucket 76: [Remember When? - Roger Varley the Sweep]
B077	Sanam Aap Ki Khatir	Bucket 77: [Roger Varley the Sweep - Sanam Aap Ki Khatir]
B078	Seemabaddha	Bucket 78: [Sanam Aap Ki Khatir - Seemabaddha]
B079	Shirobara wa sakedo	Bucket 79: [Seemabaddha - Shirobara wa sakedo]
B080	Small Gods	Bucket 80: [Shirobara wa sakedo - Small Gods]
B081	Spank Bank	Bucket 81: [Small Gods - Spank Bank]
B082	Streli nad vodata	Bucket 82: [Spank Bank - Streli nad vodata]

B083	Taiketsu! Test de go!!	Bucket 83: [Streli nad vodata - Taiketsu! Test de go!!]
B084	The 38th Annual Grammy Awards	Bucket 84: [Taiketsu! Test de go!! - The 38th Annual Grammy Awards]
B085	The Bravery: Live from La Zona Rosa	Bucket 85: [The 38th Annual Grammy Awards - The Bravery: Live from La Zona Rosa]
B086	The Death of Tempo	Bucket 86: [The Bravery: Live from La Zona Rosa - The Death of Tempo]
B087	The God in the Garden	Bucket 87: [The Death of Tempo - The God in the Garden]
B088	The Land of Adventure	Bucket 88: [The God in the Garden - The Land of Adventure]
B089	The Myth of Color	Bucket 89: [The Land of Adventure - The Myth of Color]
B090	The Red Eyed Creature	Bucket 90: [The Myth of Color - The Red Eyed Creature]
B091	The Substitute Teacher	Bucket 91: [The Red Eyed Creature - The Substitute Teacher]
B092	The Wordsmith	Bucket 92: [The Substitute Teacher - The Wordsmith]
B093	Tom	Bucket 93: [The Wordsmith - Tom]
B094	Trouble	Bucket 94: [Tom - Trouble]
B095	Underwater Station	Bucket 95: [Trouble - Underwater Station]
B096	Veninder på trods	Bucket 96: [Underwater Station - Veninder på trods]
B097	Warm Steel Loft	Bucket 97: [Veninder på trods - Warm Steel Loft]
B098	Who, Me?	Bucket 98: [Warm Steel Loft - Who, Me?]
B099	Yalanci	Bucket 99: [Who, Me? - Yalanci]
B100	Þegar Kanínur Fljúga	Bucket 100: [Yalanci - Þegar Kanínur Fljúga]

3.1.6 6. Histogram Visualization

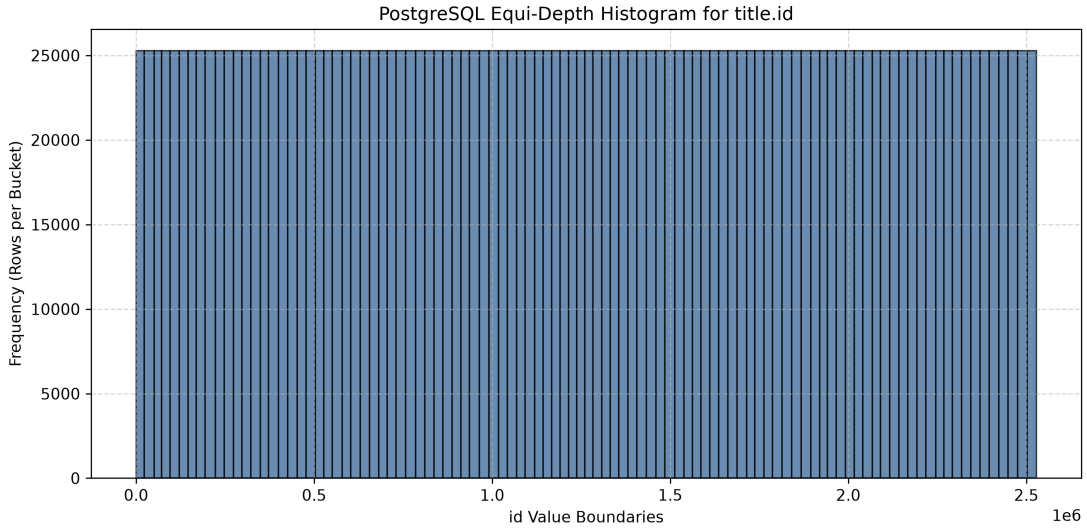


Figure 1: Histogram for title.id

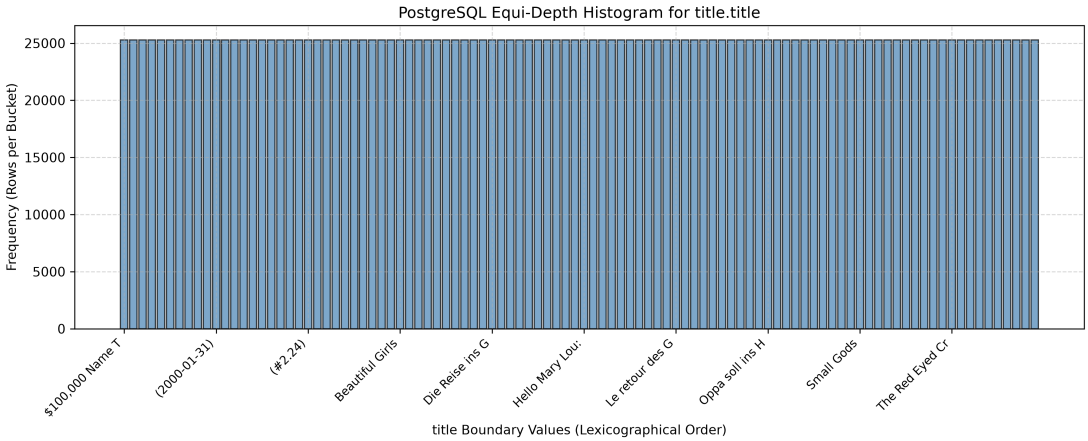


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error

- **Max Selectivity Error**: Since `default_statistics_target` is 100, there are 100 buckets. Therefore, the maximum selectivity error for a range query is bounded by:

$$\text{Max Error} \leq \frac{1}{B} = \frac{1}{100} = 0.01 \quad (1.0\%)$$

For evaluated queries(q9), the observed maximum selectivity error is **0.8147%** (0.008147), which strictly satisfies Max Error $\leq 1.0\%$.

3.1.8 8: Selectivity Estimation for Text Columns

Selectivity estimate is done for range query on text column such as "title" in PSQL such as:

*The engine scans the `histogram_bounds` array stored under pg_stats to locate the specific bucket where the target string is, after that it counts all complete buckets preceding the target bucket. Each full bucket represents a fixed fraction of total rows (1/B). For the target bucket, PostgreSQL converts initial string bytes into floating-point numbers in $[0.0, 1.0]$ range for interpolation as explain in part 4, assuming uniform distribution inside the target bucket, it calculates the target's position using:

$$\text{Fraction} = \frac{\text{Query Value} - \text{Bucket Min}}{\text{Bucket Max} - \text{Bucket Min}}$$

- Final Calculation is done by adding the fractional bucket estimate to the sum of preceding full buckets to get the final estimated selectivity.

3.1.9 9: Estimated vs. Actual Selectivities

To calculate estimated and actual selectivities of the queries and moreover there accuracy we compare estimated rows (from `EXPLAIN`) against actual rows (from `EXPLAIN ANALYZE`).

Total row count (N) for the `title` table is 2,528,312.

Query 1: `title < 'Race'`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.title < 'Race';
```

- **Actual Selectivity:** The query physically returned 1,918,141 matching rows: $S_{\text{act}} = \frac{1918141}{2528312} \approx 0.7587$ (75.87%).

```
EXPLAIN SELECT * FROM title WHERE title.title < 'Race';
```

- **Estimated Selectivity:** The planner estimated 1,938,739 rows: $S_{\text{est}} = \frac{1938739}{2528312} \approx 0.7668$ (76.68%).

- **Observation:** The absolute error is roughly 0.81%.

Query 2: `id < 350000`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.id < '350000';
```

- **Actual Selectivity:** The query returned 349,999 matching rows: $S_{\text{act}} = \frac{349999}{2528312} \approx 0.1384$ (13.84%).

```
EXPLAIN SELECT * FROM title WHERE title.id < '350000';
```

- **Estimated Selectivity:** The planner estimated 355,451 rows: $S_{\text{est}} = \frac{355451}{2528312} \approx 0.1406$ (14.06%).
- **Observation:** The absolute error here is about 0.22%.

3.2 Q2. Optimal Serial Histogram Construction ($B = 20$ Buckets)

3.2.1 1. Histogram Visualizations for Exact $N = 3000$ Samples (Slide-Faithful Horizontal Bar Plots)

As defined in lecture slides, a Serial Histogram sorts domain values by their ordered frequencies ($f(v_{(1)}) \leq f(v_{(2)}) \leq \cdots \leq f(v_{(m)})$) and partitions the frequency space to minimize frequency variance $\sum n_i V_i$. The horizontal bar charts below plot Frequencies on the X-axis and Value Set / Buckets on the Y-axis:

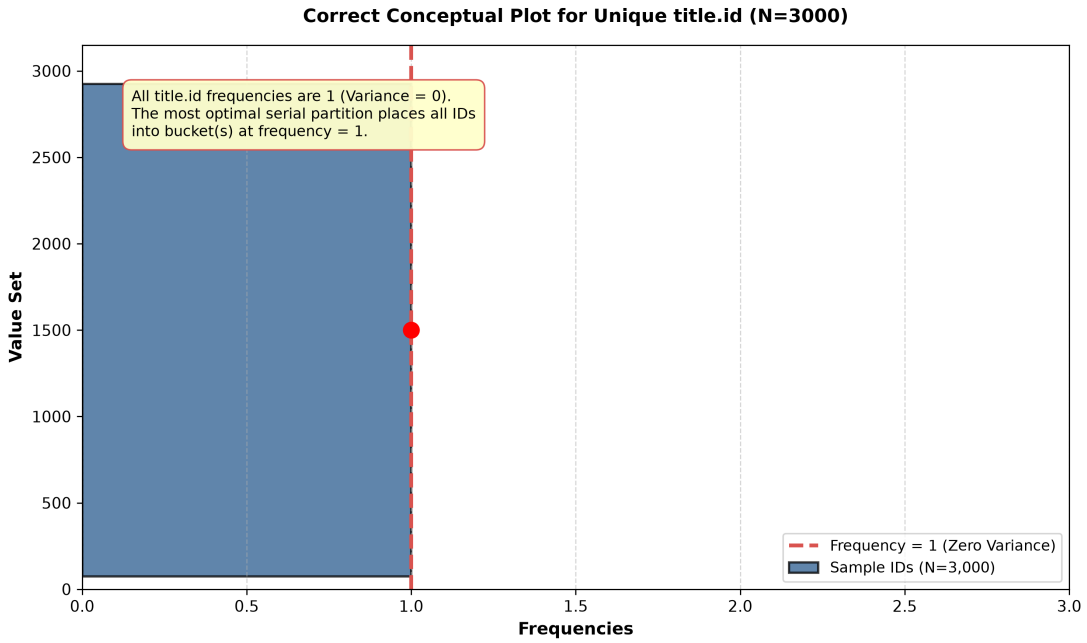


Figure 3: Correct Conceptual Plot for Unique title.id (N=3000)

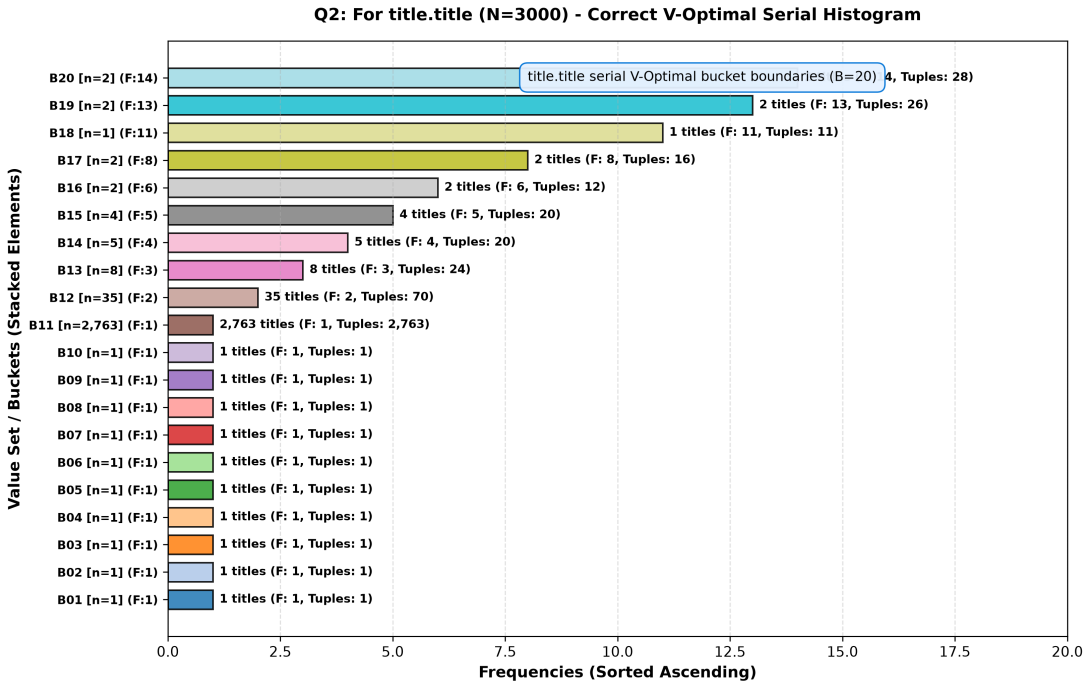


Figure 4: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

3.2.2 2. Execution Build Time (Average over 3 Runs)

- **Average Build Time for $N = 3000$: 1.2431 seconds (`id` column) / 1.2663 seconds** (`title` column)

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

The optimal serial histogram algorithm partitions the exact 3,000 samples into 20 dynamic frequency buckets. The exact bucket value ranges, distinct domain value counts (n_i), and frequency distributions for both columns are listed below:

A. Column `id` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	Value Range (`id`)	Distinct Values (n_i)	Frequency Range	Average Frequency ($\bar{f}_i$)
Bucket 01	[1,234 - 1,234]	1	1	1.00
Bucket 02	[1,377 - 1,377]	1	1	1.00
Bucket 03	[1,848 - 1,848]	1	1	1.00
Bucket 04	[4,038 - 4,038]	1	1	1.00
Bucket 05	[4,148 - 4,148]	1	1	1.00
Bucket 06	[4,770 - 4,770]	1	1	1.00
Bucket 07	[5,046 - 5,046]	1	1	1.00
Bucket 08	[5,703 - 5,703]	1	1	1.00
Bucket 09	[7,122 - 7,122]	1	1	1.00
Bucket 10	[7,248 - 7,248]	1	1	1.00
Bucket 11	[7,425 - 7,425]	1	1	1.00
Bucket 12	[7,557 - 7,557]	1	1	1.00
Bucket 13	[7,662 - 7,662]	1	1	1.00
Bucket 14	[8,124 - 8,124]	1	1	1.00
Bucket 15	[8,547 - 8,547]	1	1	1.00
Bucket 16	[8,598 - 8,598]	1	1	1.00
Bucket 17	[9,616 - 9,616]	1	1	1.00
Bucket 18	[10,047 - 10,047]	1	1	1.00
Bucket 19	[10,771 - 10,771]	1	1	1.00

Bucket 20	[10,948 - 2,528,274]	2981	1	1.00

B. Column `title` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 20$)

Bucket #	Value Range (`title`)	Distinct Values (n_i)	Frequency Range	Average Frequency ($\bar{f}_i$)
Bucket 01	["\$#!+ Happens" ... "\$#!+ Happens"]	1	1	1.00
Bucket 02	["'42 Woodie Part" ... "'42 Woodie Part"]	1	1	1.00
Bucket 03	["(#1.100)" ... "(#1.100)"]	1	1	1.00
Bucket 04	["(#1.10033)" ... "(#1.10033)"]	1	1	1.00
Bucket 05	["(#1.10271)" ... "(#1.10271)"]	1	1	1.00
Bucket 06	["(#1.1043)" ... "(#1.1043)"]	1	1	1.00
Bucket 07	["(#1.104)" ... "(#1.104)"]	1	1	1.00
Bucket 08	["(#1.10594)" ... "(#1.10594)"]	1	1	1.00
Bucket 09	["(#1.10693)" ... "(#1.10693)"]	1	1	1.00
Bucket 10	["(#1.106)" ... "(#1.106)"]	1	1	1.00
Bucket 11	["(#1.10666)" ... "Ônna no kaôyaku"]	2763	1	1.00
Bucket 12	["(#1.132)" ... "The Box"]	35	2	2.00
Bucket 13	["(#1.118)" ... "(#4.1)"]	8	3	3.00
Bucket 14	["(#1.11)" ... "(#3.1)"]	5	4	4.00
Bucket 15	["(#1.10)" ... "Pilot"]	4	5	5.00
Bucket 16	["(#1.7)" ... "(#1.9)"]	2	6	6.00
Bucket 17	["(#1.6)" ... "(#1.8)"]	2	8	8.00
Bucket 18	["(#1.5)" ... "(#1.5)"]	1	11	11.00
Bucket 19	["(#1.1)" ... "(#1.2)"]	2	13	13.00
Bucket 20	["(#1.3)" ... "(#1.4)"]	2	14	14.00

3.2.4 4. 3-Run Selectivity Error Calculation & Average ($N = 3000$)

As required to account for sample randomness, the selectivity error evaluation was executed across **3 independent runs** (with exact sample size $N = 3000$). All three run errors are listed below along with their computed average:

A. 3-Run Max Selectivity Error Summary ($N = 3000$)

Column	Run 1 Max Error	Run 2 Max Error	Run 3 Max Error	Average Max Selectivity Error
id	0.023000	0.007895	0.010420	0.013772 (1.3772%)
title	0.624589	0.609475	0.617858	0.617307 (61.7307%)

B. Quantile-by-Quantile Error Evaluation Table for Run 1 ($N = 3000, B = 20$)

Index (k)	Query Threshold (q_k)	Actual Count (A)	Actual Selectivity (S_{act})	Estimated Count (E)	Estimated Selectivity (S_{est})	Absolute Error $\lvert S_{\text{est}} - S_{\text{act}} \rvert$
q_0	1,234.0	1	0.000333	1.00	0.000333	0.000000
q_1	282,016.2	352	0.117333	340.00	0.113332	0.004001
q_2	562,798.4	673	0.224333	672.50	0.224166	0.000168
q_3	843,580.7	966	0.322000	1005.00	0.334999	0.012999
q_4	1,124,362.9	1307	0.435667	1337.50	0.445833	0.010166
q_5	1,405,145.1	1613	0.537667	1670.00	0.556666	0.019000
q_6	1,685,927.3	1946	0.648667	2002.50	0.667500	0.018833
q_7	1,966,709.6	2266	0.755333	2335.00	0.778333	**0.023000**
q_8	2,247,491.8	2623	0.874333	2667.50	0.889167	0.014833
q_9	2,528,274.0	3000	1.000000	3000.00	1.000000	0.000000

3.2.5 5. Histograms Across Exact Sample Sizes ($N = 1000, 3000, 5000$)

Below are the optimal serial histogram visualizations ($B = 20$) for exact sample sizes $N = 1000$, $N = 3000$, and $N = 5000$:

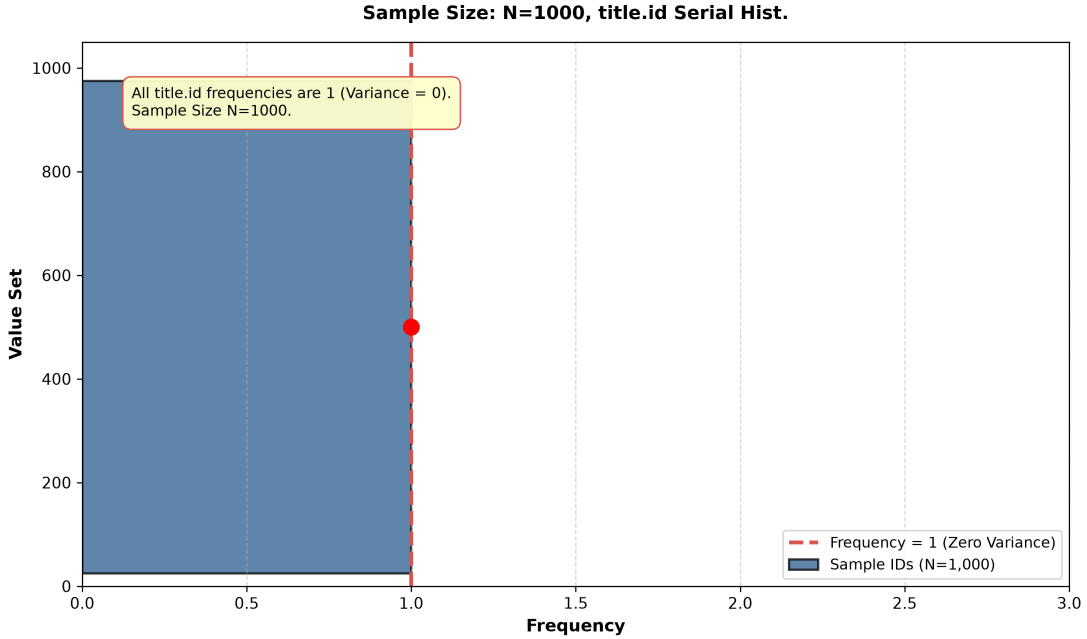


Figure 5: Sample Size: N=1000, title.id Serial Hist.

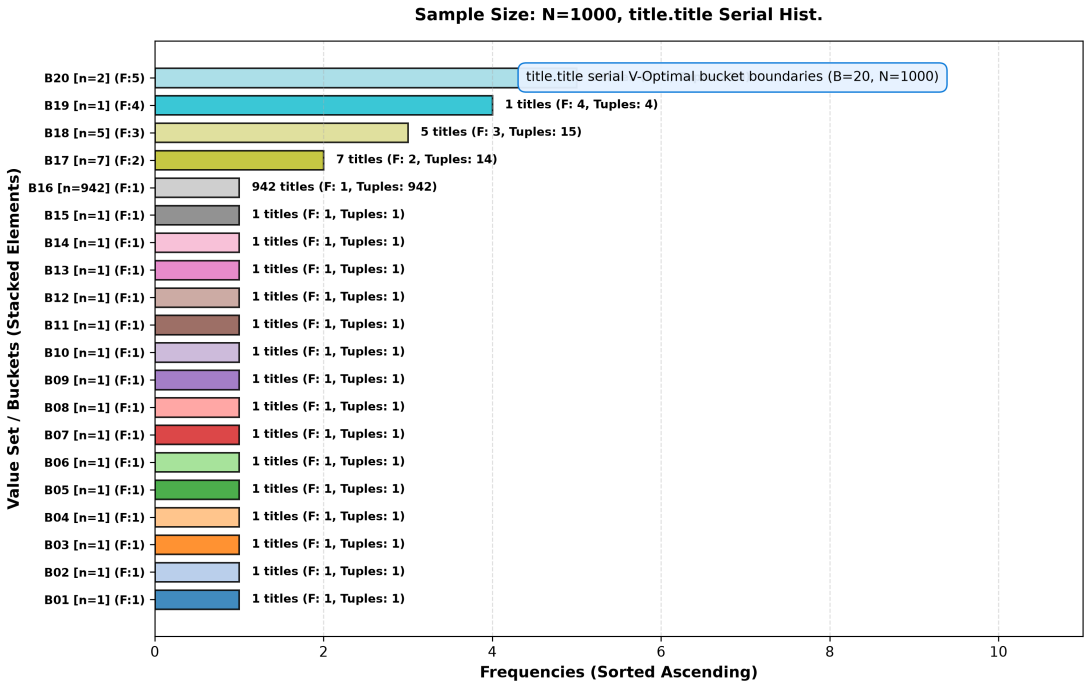


Figure 6: Sample Size: N=1000, title.title Serial Hist.

A. Sample Size $N = 1000$ Histograms

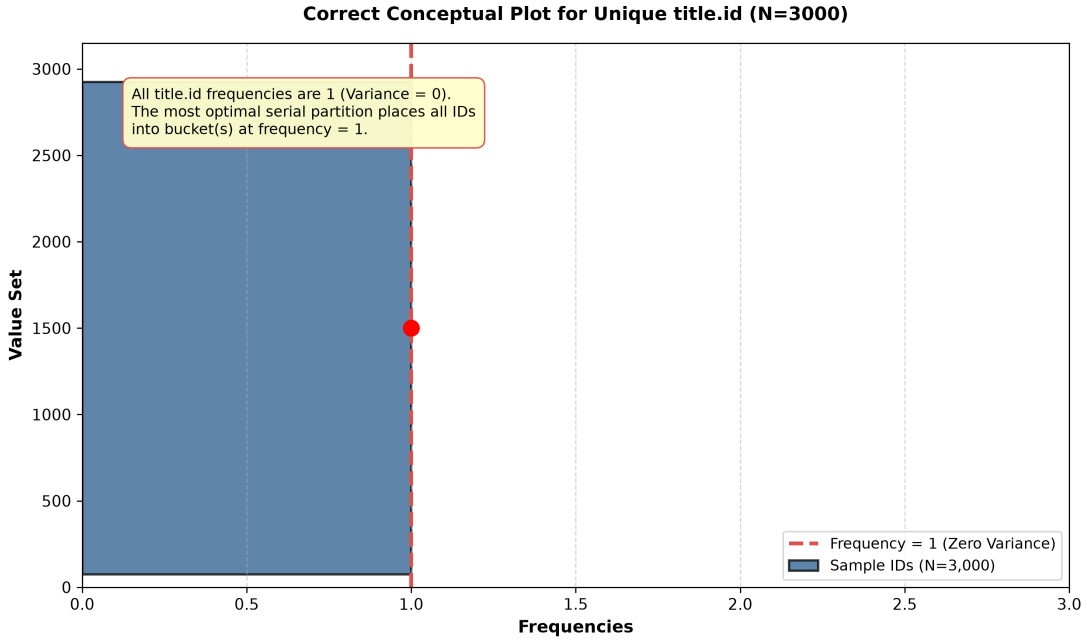


Figure 7: Correct Conceptual Plot for Unique title.id (N=3000)

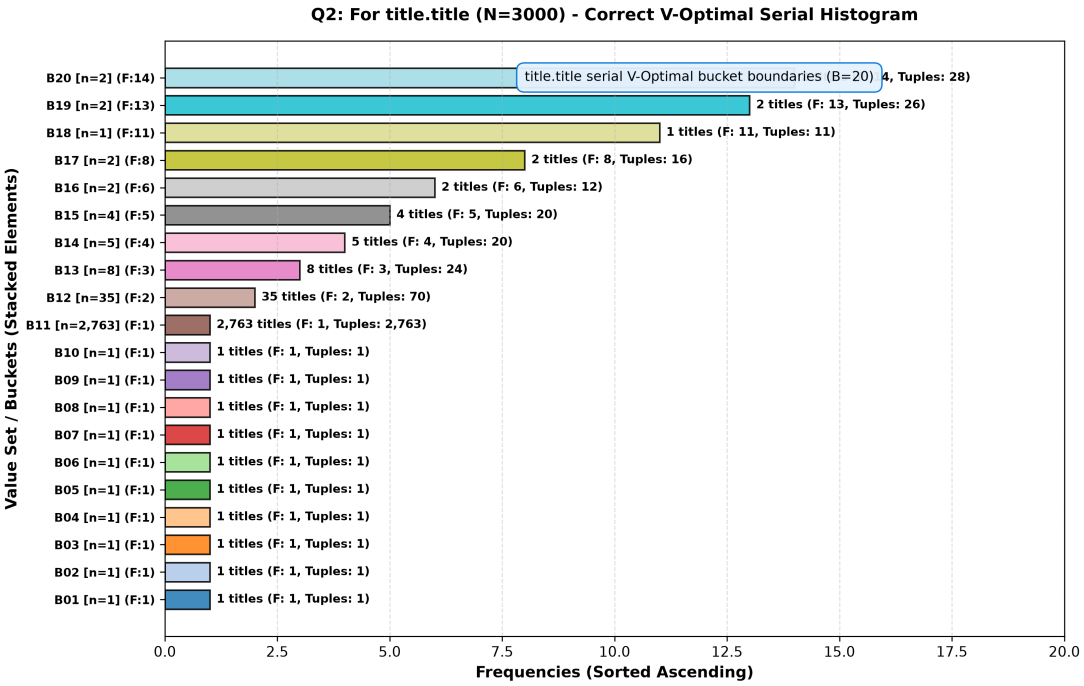


Figure 8: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

B. Sample Size $N = 3000$ Histograms

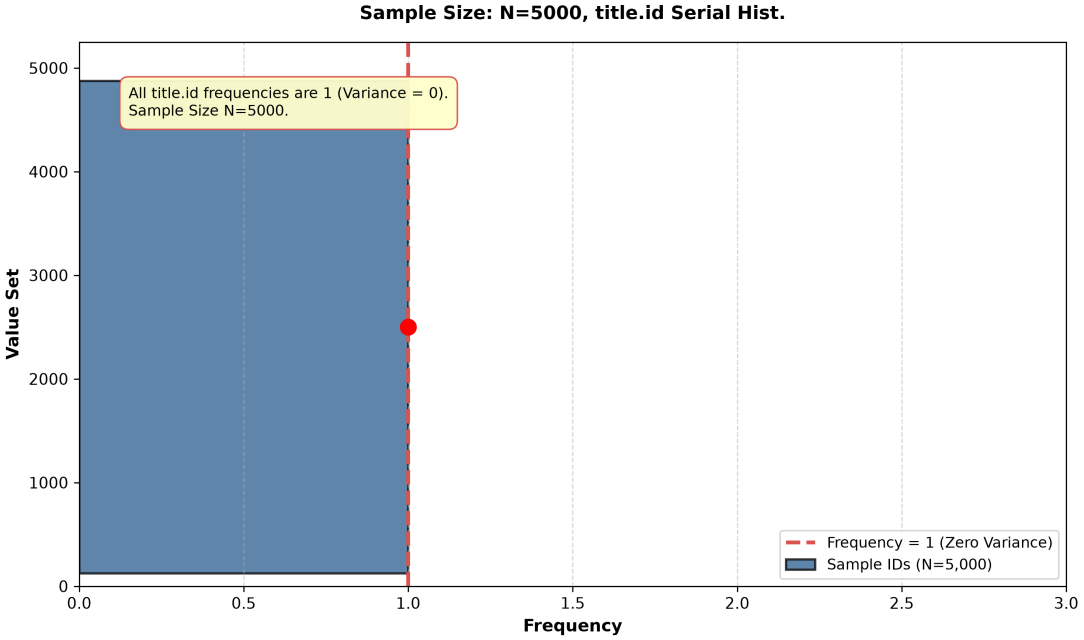


Figure 9: Sample Size: N=5000, title.id Serial Hist.

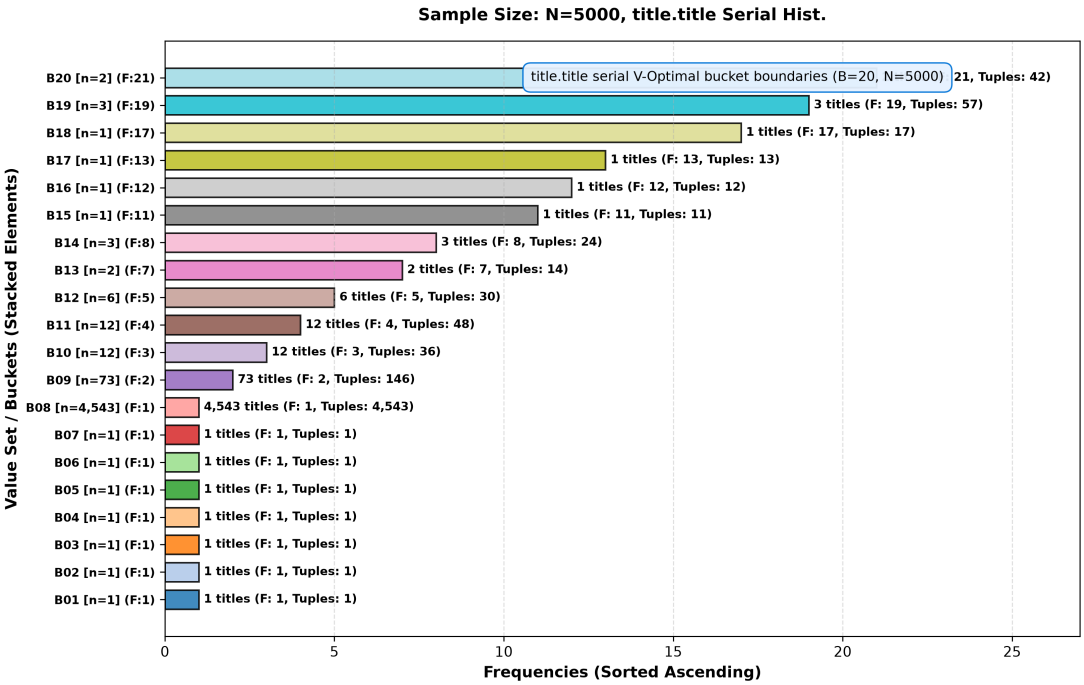


Figure 10: Sample Size: N=5000, title.title Serial Hist.

C. Sample Size $N = 5000$ Histograms

D. Comprehensive 3-Run Summary Table Across Exact Sample Sizes

Exact Sample Size (N)	Column	Run 1 Error	Run 2 Error	Run 3 Error	Average Max Error	Average Build Time
1000	id	0.021797	0.013345	0.012868	0.016003	0.4297s
1000	title	0.630636	0.618118	0.167797	0.472183	0.5066s
3000	id	0.023000	0.007895	0.010420	0.013772	1.2431s
3000	title	0.624589	0.609475	0.617858	0.617307	1.2663s
5000	id	0.013780	0.005044	0.007486	0.008770	2.5777s
5000	title	0.616176	0.613055	0.608567	0.612600	2.4402s

3.2.6 6. Full Step-by-Step Linear Regression Extrapolation ($N = 2,528,312$)

To extrapolate the construction build time for the entire `title` table ($N = 2,528,312$), we fit a linear regression model on the 3-run average measured build times:

$$T(N) = m \cdot N + c$$

A. Sample Data Points (N_i, T_i)

- Point 1: $(N_1, T_1) = (1000, 0.4297)$
- Point 2: $(N_2, T_2) = (3000, 1.2431)$
- Point 3: $(N_3, T_3) = (5000, 2.5777)$

B. Mean Calculations

- $\bar{N} = \frac{1000+3000+5000}{3} = \mathbf{3000}$
- $\bar{T} = \frac{0.4297+1.2431+2.5777}{3} = \mathbf{1.4168}$ seconds

C. Deviations & Sum of Products

N_i	T_i (s)	$(N_i - \bar{N})$	$(T_i - \bar{T})$	$(N_i - \bar{N})^2$	$(N_i - \bar{N})(T_i - \bar{T})$
1000	0.4297	-2000	-0.9871	4,000,000	1974.2404
3000	1.2431	0	-0.1737	0	-0.0000
5000	2.5777	+2000	1.1609	4,000,000	2321.7339
Sum				**8,000,000**	**4295.9743**

D. Slope (m) and Intercept (c) Formulas

- **Slope (m)**:

$$m = \frac{\sum(N_i - \bar{N})(T_i - \bar{T})}{\sum(N_i - \bar{N})^2} = \frac{4295.9743}{8,000,000} = \mathbf{0.00053700} \text{ seconds / sample}$$

- **Intercept (c)**:

$$c = \bar{T} - m \cdot \bar{N} = 1.4168 - (0.00053700 \times 3000) = -\mathbf{0.194165} \text{ seconds}$$

E. Extrapolated Build Time for Full Table ($N = 2,528,312$)

$$T(2,528,312) = (0.00053700 \times 2,528,312) + (-0.194165) = \mathbf{1357.50} \text{ seconds} \quad (\mathbf{22.63} \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

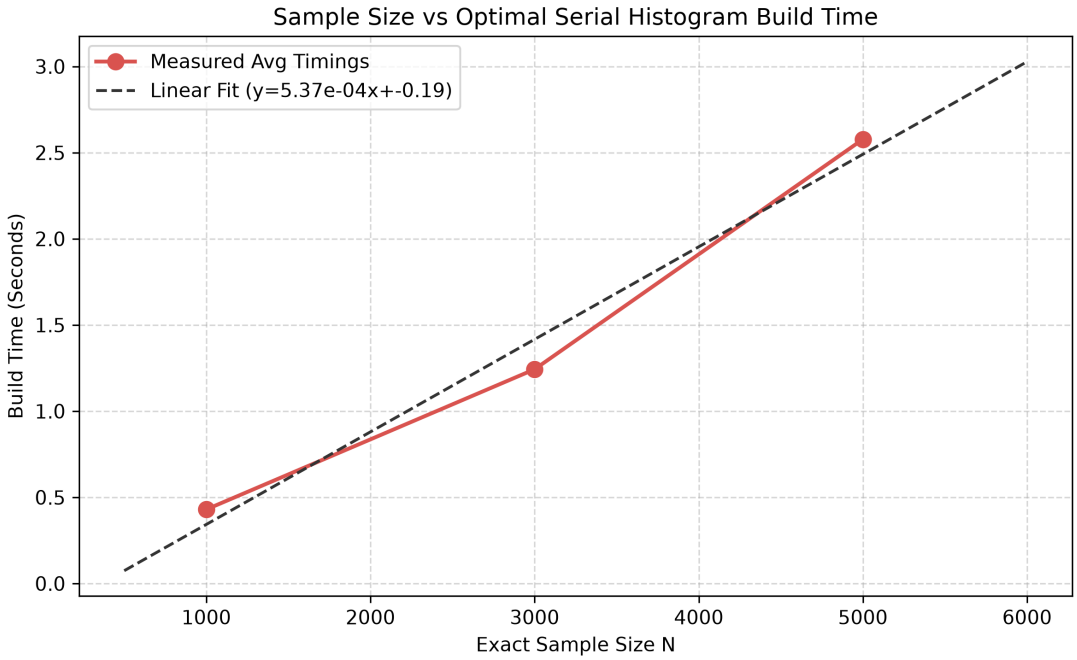


Figure 11: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

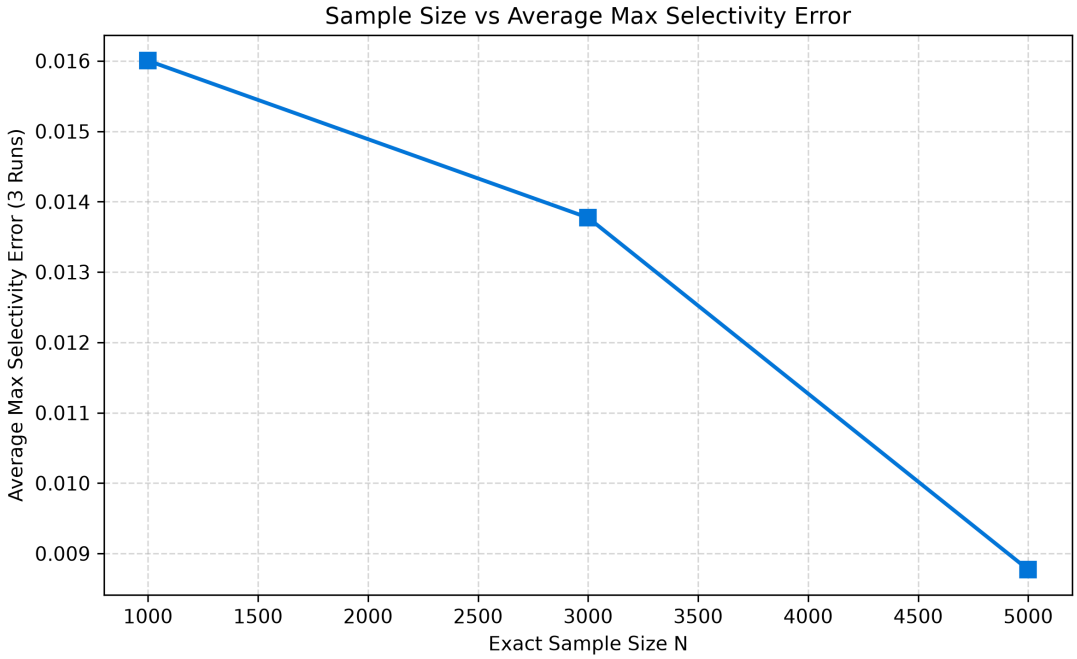


Figure 12: Sample Size vs Max Selectivity Error

3.3 9: The Trade-off of Bucket Counts

When building a histogram, choosing the number of buckets (B) is a classic trade off between performance and accuracy.

Choosing Fewer Buckets (e.g., 5 to 10)

If we use a small number of buckets, the biggest advantage is speed and efficiency. The histogram takes up very little memory in the system catalog and builds much faster because the time complexity of our algorithm, $O(B \cdot N^2)$, drops significantly. The query planner also spends less time analyzing it. However, the major downside is that we lose a lot of data resolution. Because the buckets are so wide, the internal mathematical estimates become much less accurate, especially if the data is heavily skewed or unevenly distributed.

Choosing More Buckets (e.g., 50 to 100)

On the flip side, using a large number of buckets gives the database a high-definition view of how the data is distributed. The estimates become incredibly precise, which leads to much lower selectivity errors and helps the query optimizer make smarter decisions for complex joins. The catch is that this accuracy comes at a heavy system cost. It takes noticeably longer to calculate and construct the histogram, consumes more storage space, and adds a slight delay to the query planning phase before execution can actually begin.